THE COMPOSITION OF THE CYCLE FAMILIES ASSOCIATED TO COHERENT SHEAVES IS THE FAMILY ASSOCIATED TO THE TENSOR PRODUCT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Cycle family associated to a coherent sheaf (relative-cycles chapter, Example example-family-associated-module, canonical tag 0H4Q). cited at statement line 2848. Defines the family-associated-module construction producing alpha and beta in the statement.
- Coherence of the bivariant action (relative-cycles chapter, Lemma lemma-action-coherent, canonical tag 0H69). cited at proof line 2865. Computes the fibre of the composition as the cycle of the tensor of the fibres for flat F_s.

Source reference. Original reference 0H4Q: sources/stacks-chow/tags.txt, source lines 21360–21360. Cited at statement line 2848: defines the construction producing alpha and beta.

Source reference. Original lemma 0H69: sources/stacks-chow/tags.txt, source lines 21414–21414. Cited at proof line 2865: computes the fibre of the composition.

Lemma 0.2. *Let $f : X \to Y$ and $Y \to S$ be morphisms of schemes, both locally of finite type. Let $r, e \geq 0$. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module of finite type, with $\dim(\mathit{Supp}(\mathcal{F}_y)) \leq r$ for all $y \in Y$. Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_Y$-module of finite type, with $\dim(\mathit{Supp}(\mathcal{G}_s)) \leq e$ for all $s \in S$. If $\alpha = [\mathcal{F}/X/Y]_r$ and $\beta = [\mathcal{G}/Y/S]_e$ (Example 0H4Q) and $\mathcal{F}$ is flat over Y, then $\alpha \circ \beta = [\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}/X/S]_{r+e}$.*

Proof. First we observe that $\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}$ is a quasi-coherent $\mathcal{O}_X$-module of finite type. Let $s \in S$. Observe that

$$(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G})_s = \mathcal{F}_s \otimes_{\mathcal{O}_{X_s}} f_s^*\mathcal{G}_s$$

by right exactness of tensor products. Moreover $\mathcal{F}_s$ is flat over Y_s as a base change of a flat module. Thus the equality $(\alpha \circ \beta)_s = [(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G})_s]_{r+e}$ follows from Lemma 0H69. □

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